

CE HOU (ALBERT)

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EDUCATION

Hong Kong University of Science and Technology

Doctor of Philosophy (PhD) in Civil Engineering

Clear Water Bay, Hong Kong

Feb 2023 – Nov 2026 (expected)

- Advisors: Prof. Fan Zhang (at Peking University) and Prof. Sen Li

University College London

Master of Science

London, United Kingdom

Sep 2021 – Sep 2022

- Major in Spatial Data Science and Visualization
- Advisor: Prof. Chen Zhong

Wuhan University

Bachelor of Engineering

Wuhan, China

Sep 2017 – Jun 2021

- Major in Geodesy and Geomatics Engineering
- Advisors: Prof. Bao Zhang and Prof. Yibin Yao

VISITING EXPERIENCE

University of California, Berkeley

Visiting PhD student

CA, United States

Mar 2026– Sep 2026 (expected)

- Advisor: Prof. Lu Liang

Peking University

Visiting PhD Student

Beijing, China

Feb 2024 – Now

- Advisor: Prof. Fan Zhang

PUBLICATIONS

Manuscripts under revision:

- Yidan Wang, Fan Zhang*, **Ce Hou**, Yingjing Huang, Yuxuan Liang, Filip Biljecki, Liyan Xu, Limin Zhang. Uncovering Historical City Connections with Satellite Imagery. *Nature Cities*
- Entong Ke, **Ce Hou**, Tianshun Gu, Quan Qin, Qinyu Cui, Haochen Shi, Tian Lan*, Yan Zhang*. The universal decay of nighttime activity reveals human rhythms across cities. *Nature Cities*
- Shizhen Wang, **Ce Hou***, Yiqi Huang, Yuhong Wang, Yan Song. Behavioral Inertia in Sustainable Urban Mobility: Consistency of Place Preferences under External Shocks. *Npj Sustainable Mobility and Transport*

Peer-reviewed papers:

- Yecheng Zhang, Rong Zhao, Zhizhou Sha, Yong Li, Lei Wang, **Ce Hou**, Wen Ji, Hao Huang, Yunshan Wan, Jian Yu, Junhao Xia, Yuru Zhang, Chunlei Shi (2026). UrbanAlign: Post-hoc Semantic Calibration for VLM-Human Preference Alignment. *2026 European Conference on Computer Vision (ECCV)*. <https://arxiv.org/abs/2602.19442>
- Maoran Sun, **Ce Hou**, Qiaosi Li, Fan Zhang, Ronita Bardhan, Qunshan Zhao* (2025) Deciphering Exterior: Building Energy Efficiency Prediction with Emerging Urban Big Data. *npj Urban Sustainability*. <https://doi.org/10.1038/s42949-026-00348-7>
- Qianbao Hou, **Ce Hou**, Fan Zhang, Qihao Weng (2025). Multi-source geo-localization in urban built environments for crowd-sourced images by contrastive learning. *ISPRS Journal of Photogrammetry and Remote Sensing*. <https://doi.org/10.1016/j.isprsjprs.2025.09.024>
- **Ce Hou**, Fan Zhang*, Yong Li, Haifeng Li, Gengchen Mai, Yuhao Kang, Ling Yao, Wenhao Yu, Yao Yao, Song Gao, Min Chen, Yu Liu (2025). Urban Sensing in the Era of Large Language Models. *The innovation*. 6(1). DOI: <https://doi.org/10.1016/j.xinn.2024.100749>

- **Ce Hou**, Yuhao Kang, Fan Zhang*, Song Gao, Yong Li, Fabio Duarte, Sen Li (2024). “Transferred bias” uncovers the balance between the development of physical and socio-economic environments of cities. *Annals of the American Association of Geographers*. <https://doi.org/10.1080/24694452.2024.2412173>
- Lei Wang, **Ce Hou**, Yecheng Zhang, Jie He* (2024). Measuring solar radiation and spatio-temporal distribution in different street network direction through solar trajectories and street view images. *International Journal of Applied Earth Observation and Geoinformation*. <https://doi.org/10.1016/j.jag.2024.104058>
- Yikang Wang, Yuhao Kang, Haokun Liu, **Ce Hou**, Bing Zhou, Shan Ye, Yuyan Liu, Jinmeng Rao, Zhenghao Pei, Xiang Ye, Song Gao (2023). Choosing GIS graduate programs from afar: Chinese students’ perspectives. *Transactions in GIS*. DOI: <https://doi.org/10.1111/tgis.13037>

Preprints:

- Furong Jia, Lanxin Liu, **Ce Hou**, Fan Zhang*, Xinyan Liu, Yu Liu (2025). Towards Interpretable Geo-localization: a Concept-Aware Global Image-GPS Alignment Framework. *arXiv preprint arXiv:2509.01910*.
- **Ce Hou**, Fan Zhang*, Yuhao Kang, Zhuangyuan Fan, Sen Li (2025). Transferring population group knowledge from multimodal large language model to small model: using urban safety perception evaluation as case study, 07 May 2025, PREPRINT (Version 1) available at Research Square [<https://doi.org/10.21203/rs.3.rs-6351052/v1>]
- Piyu Ke, Xiaofan Gui, Wei Cao, Dezhi Wang, **Ce Hou**, Lixing Wang, Xuanren Song, Yun Li, Biqing Zhu, Jiang Bian, Stephen Sitch, Philippe Ciais, Pierre Friedlingstein, and Zhu Liu* (2023). Near-real-time monitoring of global ocean carbon sink. *arXiv preprint arXiv:2312.01637*, 2023.

Book chapters:

- **Ce Hou**, Yong Li, Fan Zhang* (2024). Sensing Urban Physical Environment with GeoAI and Street-Level Imagery. *Handbook of Geospatial Approaches to Sustainable Cities*.

RESEARCH EXPERIENCE

Tsinghua University

Research Assistant

Near-real-time Global Carbon Budget estimation (ANGCB)

- Advisors: Prof. Zhu Liu

Beijing, China

Jul 2022 – Dec 2022

The Bartlett School of Sustainable Construction, UCL

Research Assistant

Extreme risks of energy demand insecurity for space heating and cooling in the United Kingdom

- Advisors: Prof. Zhifu Mi and Dr. Haoran Zhang

London, United Kingdom

May 2022 – May 2023

GRANTS

- **Co-investigator**, Housing conditions and long-term outcomes in the UK, funded by *British Academy*, 9,500 GBP
May 2026 – May 2028

INVITED TALKS

Transferred bias uncovers the balance between the development of physical and socio-economic environments of cities

Invited talk at School of Public Administration, China University of Geosciences (Wuhan)

Wuhan, China

Mar 2025

Leveraging GeoAI and LLM for sensing the sustainable urban development

Invited talk at Department of Geography and the Environment, University of Texas at Austin

Texas, United States

Mar 2026

Street view imagery as a lens for sensing buildings

California, United States

Invited talk at Center for the Built Environment, UC Berkeley

Apr 2025

TEACHING EXPERIENCE

Geospatial Science & Technology for Smart City (CIVL4100U)

Teaching assistant

Hong Kong SAR, China

2023-24 Fall

Introduction to Smart City Economics (CIVL4640)

Teaching assistant

Hong Kong SAR, China

2024-25 Spring

HONORS

- Best Student Paper Award, 33rd International Conference on Geoinformatics 2026
- Overseas Research Award, HKUST 2025
- Best Poster Award, the 2nd Research Summit of Urban Science and Human Dynamics 2024
- Creativity Award, Research Summit of Urban Science and Human Dynamics 2023
- Bezos/Musk Prize for Entrepreneurial Potential in Knight Frank Dubai Housing Data Hackathon 2022
- Excellent undergraduate graduation design, Wuhan University 2021
- Distinct Graduate Student, Wuhan University 2021
- Distinct Student, Wuhan University 2021
- Li Qinghai Scholarship for Outstanding Students in Geomatics 2021
- First Class Scholarship, Wuhan University 2021
- Distinct Student, Wuhan University 2020
- Outstanding student leaders 2020
- Second Class Scholarship, Wuhan University 2020
- Social activist Student, Wuhan University 2020
- Merit Student, Wuhan University 2019

COMMUNITY SERVICES

The Chinese Professional in Geographic Information Systems

BOD student member

Fulfilling board responsibilities and participate in CPGIS's management and development.

Online

Jan 2026 – Now

GISphere

Co-chair

Participate in organizing online seminars (GISalon) and people interview section (GISstory), assisting in managing the official WeChat account and publishing tweets.

Online

Mar 2022 – Mar 2024

The Chinese Professional in Geographic Information Systems

Volunteer

Assist in managing the official WeChat account and publishing tweets.

Online

Dec 2021 – Now

Journal reviews:

Scientific Data; Computer, Environment and Urban System; International journal of Geographical Information Science; Humanity and Social Science Communications; Information Geography; Transactions in GIS; Scientific Reports; Journal of Urban Technology; Journal of Asian Architecture and Building Engineering; Computational Urban Science